Supplementary material

Does the choice of Sleep Regularity Index calculator or sleep/wake input change the conclusions?

Reference numbers refer to the reference list of the main text. Table and figure numbers without the prefix S refer to the main text.

1 RIRI reporting items

Table S1: RIRI reporting items [20] (Table 5) for the present re-analysis.

#	Item	This study
Methods		
1	Data collection process	NHANES 2011–2014, wrist-worn ActiGraph GT3X+ worn 24 h for up to 9 days. NCHS released a minute-level wake/sleep/non-wear/unknown classification (`PAXPREDM`). No sleep diaries. The raw 80 Hz signal of the same recordings was processed again with `GGIR` and the public SWaN code (main text Sections 2.1 and 2.4).
2	Performance of measure	`PAXPREDM` classifier [10]: F1 0.84 (sleep-wear vs wake-wear) and 0.94 (wake-wear vs non-wear) on independent validation sets against polysomnography and diaries. Epoch-level fragmentation of the released series is reported in main text Table 1.
3	Duration of recording interval	3 to 7 complete calendar days per participant after the published filters (median 7, 60% with seven consecutive days, main text Section 2.1).
4	Frequency of measure	80-Hz raw acceleration summarised by NCHS to 1-minute epochs; SRI computed at 1-minute epochs (`sleepreg`: re-expressed at 10 s; `GGIR`: 1-min epochs as supplied, since the function aggregates only epochs shorter than 30 s). Raw signal: 5-s epochs (`GGIR`) and 30-s windows (SWaN), aggregated to 1-minute labels. No other down-sampling (Table S2).
5	Data processing tool, if applicable	Released minute classification (`PAXPREDM`) with four smoothing variants (rolling majority; `GGIR` SRI1 rule; bridging of awakenings $<$30 min and removal of isolated sleep $<$30 min; main sleep period only). Raw signal, in a random sample of 1000 participants (main text Section 2.4): `GGIR` 3.3-6 (autocalibration, ENMO and z-angle, default non-wear detection, sleep period by HDCZA, sustained inactivity bouts by the van Hees rule) and `swan_accel` 1.18 (SWaN); settings in Table S2; code shared.

Table S1, continued

#	Item	This study
6	Quality control	Published filters: full-minute epochs, no quality flag, ≥ 3 days with < 120 min non-wear, days of exactly 1440 minutes, ≥ 3 consecutive days incl. a weekend day, $< 30\%$ missing minutes. Excluded minutes and days carried as missing in one continuous timeline from monitor placement. `GGIR` day pairs kept only with $\geq 66\%$ valid epoch pairs. `sleepreg` day exclusion (> 6 h missing) and minimum comparison days (5 and 3) as per package (main text Section 2.1, Table S2).
7	Mathematical formula	$\mathrm{SRI} = -100 + \frac{200}{N_v} \sum 1[s_i = s_{i+1440}]$ over valid epoch pairs 24 h apart [1], days paired by weekday index in the published implementation (main text Section 2.2).
8	Regularity index calculator, if applicable	Custom R code (published implementation), `GGIR` 3.3-6 `CalcSleepRegularityIndex` (SRI1 engine, smoothing off), `sleepreg` 1.3.5 `SRI_from_binary` run from source, and `pyActigraphy` `sri()` for verification (Table S2).
Results		
9	Participant flow and data collection	Participants with minute files: 6917 (2011–2012) and 7776 (2013–2014). Remaining after each filter (2011–2012 / 2013–2014): full-minute epochs 6917 / 7776, no quality flag 6916 / 7773, non-wear rule 6341 / 6903, 1440-minute days 6291 / 6832, consecutive days with a weekend day 5459 / 5882, missingness 5243 / 5562. Analytic sample after covariate availability: 7085 (published flow chart). SRI available for 7076 (formula), 7071 to 7073 (`GGIR`) and 3673 to 3854 / 5342 to 5423 (`sleepreg`, 5 / 3 comparison days), depending on the input (Section 5).
10	Percentage of data collection interval with data available	66.2 million retained minutes. About 6% of retained minutes are non-wear or unknown (1.7% and 4.2%). Of the participants, 40% have fewer than seven consecutive complete days (Section 5).
11	Descriptive statistics	Median by input variant and calculator (main text Table 2); distributions in Figure S3; by sex in the published study.
12	Comparison with previously published data	Published NHANES values [6–8], UK Biobank [3, 20] and MESA [9] discussed in the Introduction and Discussion of the main text.
Other information		
13	Additional configurations	Package defaults except: `sleepreg` `minSRIdays` 5 (default) and 3; timestamps offset by 30 s for `sleepreg`; `GGIR` smoothing options unset for the calculator comparison and re-implemented as an input variant (Table S2). Raw-data processing at package defaults except the arguments in Table S2.
14	Availability of data, code, and other materials	NHANES public files; derived data, synthetic test recordings, the minute-level series of every derivation, the container definition of the cloud runs and all code in the companion repository and the Zenodo archive (main text, Data availability).

2 Calculators, versions and invocations

Table S2: Calculators, versions and SRI-relevant settings.

Calculator	Version, call and settings
Published implementation	R code of the earlier study: Phillips formula at 1-min epochs; epoch pairs 24 h apart with both states observed; days paired by weekday index, so that a non-adjacent pair of days is left out of the comparison.
`GGIR` 3.3-6	`CalcSleepRegularityIndex(data, epochsize = 60, desiredtz = "UTC")`, the SRI1 engine, with `SRI1_smoothing_wsize_hrs` and `SRI1_smoothing_frac` unset. Input: one row per minute with columns `time`, `sleepstate`, `invalid` (1 for missing minutes). Day pairs kept if `frac_valid` ≥ 0.66, which approximates the package's own rule in `g.part4` (`frac_valid` above `includenightcrit`/24 with the default of 16 h), then averaged.
`sleepreg` 1.3.5	`SWS_from_binarySW()` then `SRI_from_binary(minSRIdays = 5 or 3)` at its default `exclNAhrs = 6` (noon-to-noon days with more than 6 h missing are excluded), run from the package source. Input: one row per minute from 00:00 of the first retained day to 24:00 of the last (the days before the first retained day are entirely missing in the shared timeline and cannot change the result), missing minutes as `NA`, timestamps offset by 30 s.
`pyActigraphy` 1.2.1	`sleep.scoring.sri.sri()` on a one-minute pandas series without missing values (verification only).
`GGIR` 3.3-6, raw signal	`GGIR(mode = 1:5)` on one CSV file per participant (80 Hz, units g, timestamps in UTC; `rmc.sf = 80`), `do.cal = TRUE`, `windowsizes = c(5, 900, 3600)`, `HASPT.algo = "HDCZA"`, `HASIB.algo = "vanHees2015"`, `sleepwindowType = "SPT"`, `includenightcrit = includedaycrit = 16`. All other arguments that affect the classification were left at the package defaults (`do.zcx = do.zcy = do.zcz = TRUE`, `do.sibreport = TRUE`, `save_ms5raw_format = "csv"` and `save_ms5raw_without_invalid = FALSE` control the output only). The 5-second epoch series of part 5 was saved (`save_ms5rawlevels = TRUE`).
SWaN, `swan_accel` 1.18	Python 3.8.20 (x86-64) with scikit-learn 0.23.2, NumPy 1.19.2, pandas 1.1.3 and SciPy 1.5.2. `swan_first_pass.main(df, file_path, sampling_rate = 80)` on each hourly file, then `swan_second_pass.main(day_folder)` on each day. Input in g with the package's column names, output at 30 s (wear, sleep, non-wear). Source: `https://github.com/binodthapachhetry/SWaN`.
Computing environment	Two laptops (Apple M5 Max and Apple M1 Pro) and x86-64 virtual machines of three cloud providers, with R 4.6.1, `GGIR` 3.3-6 and identical versions of all R packages on every platform. SWaN ran under x86-64 emulation (Rosetta 2) on the laptops, because the pinned versions of its Python packages are not available for ARM. Section 7 and Table S5 give the platforms, the computing time and the check of equality between platforms.

Table S2, continued

Calculator	Version, call and settings
Derived minute series	GGIR: majority state of the twelve 5-second epochs of a minute, a tie counted as wake, minutes with more than half of the epochs flagged as non-wear missing. GGIR SIB: sleep is class spt_sleep (inside the detected sleep period and in a sustained inactivity bout). GGIR SPT: every minute inside the detected sleep period. SWaN: sleep or wake if both 30-second labels agree, unknown otherwise; non-wear missing.
Input variants	Rolling majority: centred 61-min window (30 min on either side), sleep if $\geq 50\%$ of the observed minutes in the window are sleep. GGIR SRI1 rule: the same window, sleep if $\geq 80\%$ (a re-implementation with the example values of the GGIR documentation, 1 h and 80%, since the package's own smoothing is off by default). Bridged: wake bouts < 30 min between sleep bouts set to sleep, then sleep bouts < 30 min between wake bouts set to wake, on the observed sequence. Unknown runs ≤ 15 min enclosed by the same state take it. Main sleep period: from the bridged series, the longest sleep bout per noon-to-noon day, all other sleep set to wake.

2.1 Hand-over of the rebuilt series to the packages

The published sleepreg pipeline targets GGIR 2.0-0 and is incompatible with the current GGIR (3.3-6). The sleepreg SRI engine was therefore run directly from its GGIR-independent source functions on the binary series. sleepreg reads the clock time of a recording from its first timestamp and fails when that timestamp falls exactly on midnight. Its copy of each timeline began at the first retained day with all timestamps offset by 30 seconds, which leaves its 10-second grid and its noon-to-noon days unchanged. The days before the first retained day are entirely missing in the shared timeline and their omission cannot change the result. sleepreg computes the SRI over the continuous recording at a fixed 24 h lag, without matching by weekday, drops days with more than 6 h missing and gives no value to participants with fewer than 5 valid comparison days at its default (minSRIdays).

GGIR's CalcSleepRegularityIndex computes a Phillips SRI for each pair of consecutive calendar days, matched by time of day at 1-min epochs and using the valid epochs of each pair. It then averages the day-pair values into one SRI per person. The function assigns a value of zero (instead of a missing value) to a day pair without valid epochs. The day pairs were therefore averaged only where at least 66% of the epoch pairs were valid, which approximates the package's own criterion (frac_valid above 16/24). Averaging all day pairs without it would have given a median SRI of 43.8 on the released NHANES series and a correlation of 0.76 with the Phillips formula instead of 0.993. Since version 3.2-6 (April 2025, as noted by Czeisler et al. [20]) GGIR distinguishes two outputs. SRI1 is computed from the epoch-level sleep classification, optionally after a rolling-window smoothing. SRI2 (part 6) is computed from the detected sleep period, with awakenings shorter than SRI2_WASOmin (30 min) counted as sleep. SRI2 requires GGIR's own sleep-period detection from raw accelerometry, which NHANES does not release. Its input logic was emulated by the bridged input variant. The GGIR changelog records no change to the SRI between versions 3.3-6 and 3.3-9.

For the comparison with pyActigraphy [13] on the synthetic recordings and on 300 NHANES

recordings, the missing minutes (non-wear or unknown) were filled with the preceding state, and at the start of a recording with the following state, for all four implementations. The 300 recordings were those with the fewest missing minutes (at most 2.8% of minutes) among participants with seven consecutive retained days and a mean sleep share between 15% and 60%.

3 Reproduction of the published SRI

When the published filters were applied to the public minute files and the Phillips formula was recomputed, the resulting SRI correlated 0.96 with the published values (Figure S1). The scores of 23% of participants matched to within 0.01 points. For 59% the difference exceeded one point. The rebuilt series is the output of the archived filtering code of the earlier study: running that code unchanged on the public files returned, for every participant, the same minutes, the same missing minutes and the same retained days as the rebuild. Applying the archived pairing rule to this output returned the rebuilt SRI exactly. The published values themselves could be reproduced exactly, for all 4879 and 5657 scored participants of the two survey cycles, with two earlier states of the same code that are kept in the original project. Both states pair minute i with minute $i+1440$ in recording order and omit the Saturday-to-Sunday day pair, which the archived rule includes. The omitted pair affects 63% of participants in 2011–2012 and 90% in 2013–2014 and removes a median of 17% and 15% of their day pairs, respectively. The 2011–2012 state further required five days of data where the archived rule requires three. The 2013–2014 state had no 1440-minute rule, kept the partial days of placement and removal, and dropped participant-days with 120 min or more of non-wear instead of dropping participants. With partial days in the sequence, 87% of its participants have day pairs whose two minutes differ in clock time, for a median of 17% of their pairs. For the 7076 participants with both scores, the published SRI exceeded the rebuilt one by 0.4 points on average and the medians were 61.5 and 61.3. Of these participants, 79% fell into the same quintile. The code that reproduces the published values exactly is included in the companion repository.

4 Synthetic recordings and recording gaps

The packages were run on five deterministic synthetic recordings with known SRI (identical days, a sleep block shifted by 30 minutes on alternate days, a drift of 15 minutes per day, a nap on alternate days, night sleep alternating with day sleep), on a seeded random pattern and on a seven-day recording with realistic night-to-night shifts and naps. The deterministic recordings and the random pattern were presented complete and with one middle day retained as missing minutes. The realistic recording was presented complete, with the middle day retained as missing minutes and with that day deleted from the series. As in the main analysis, day pairs from `GGIR` were averaged only where the proportion of valid epoch pairs was at least 0.66.

On the complete recordings the Phillips formula, `GGIR` and `sleepreg` all returned the known SRI (Table S3): 100 for identical days, 91.67 for the 30-minute alternating shift, 95.83 for the 15-minute daily drift, 95.83 for the nap on alternate days, -33.33 for night sleep alternating with day sleep, and less than one point from 0 for the random pattern. The formula and `GGIR` agreed to three decimals. `sleepreg` differed by at most 0.4 points because of its 10-second grid and its noon-to-noon days. `pyActigraphy` returned the same seven values to four decimals

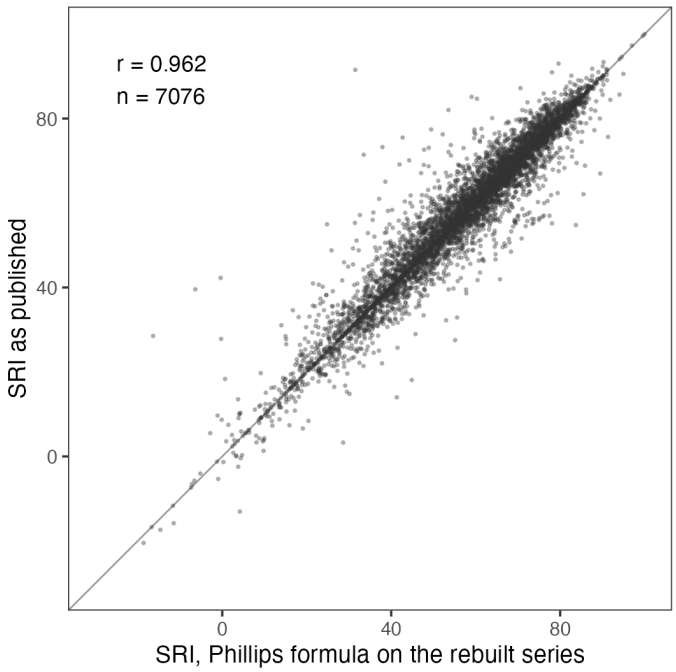


Figure S1: SRI as published in the earlier study against the Phillips formula computed on the series rebuilt from the public minute files with the published filters, for the 7076 participants with both scores. The grey line is the identity.

(`synthetic_pyact.py`). On the realistic recording all three calculators returned the same value (73.84). On the 300 NHANES recordings with the fewest missing minutes, the formula, `GGIR` and `pyActigraphy` agreed to three decimals and `sleepreg` to 0.02 points (median 71.9 for all four).

When one middle day was missing, the result depended on how the gap was presented to the package. With the day retained as missing minutes, the formula and `GGIR` agreed (68.75), whereas `sleepreg` returned no value at its default minimum number of comparison days, because a missing calendar day removes two of its noon-to-noon days (with the minimum lowered to two days it returned 66.20). With the day deleted from the series, the same recording gave 71.94 from `GGIR`, which pairs each date with the next date present and so compared two days that were 48 hours apart. `sleepreg` gave 35.18, because its transition format carries the last state before the gap over the deleted day. Neither package issued a warning. The published implementation of the formula pairs days by weekday index, skips the gap and returns 68.75 in both presentations. A further detail of `GGIR` concerns the day pairs that its inclusion criterion rejects: these carry the function's initial value of zero instead of a missing value. An unfiltered average over all day pairs (45.83 for this recording) is therefore not equivalent to the result of the package's own pipeline.

`GGIR`'s own unit test `test_CalcSleepRegularityIndex.R` builds six days at 5-second epochs with sleep blocks at varying positions and expects an SRI of 80 for day pairs 2 and 3, 100 for a constant series and -100 for a strictly increasing one. Through the wrapper used in this study the function returned 80.000 and 80.000, 100 and 100, and -100 and -100. The minute-level analogue returned 80 for every day pair, equal to the Phillips formula on each pair.

Table S3: Synthetic recordings with known SRI and a realistic seven-day recording presented complete, with day 4 retained as missing minutes, or with day 4 deleted from the series.

Recording	Known SRI	Phillips formula	`GGIR`	`sleepreg`
Identical days	100	100.00	100.00	100.00
Block shifted 30 min on alternate days	91.67	91.67	91.67	91.67
Drift of 15 min per day	95.83	95.83	95.83	95.83
30-min nap on alternate days	95.83	95.83	95.83	95.45
Night sleep and day sleep alternating	−33.33	−33.33	−33.33	−33.32
Random pattern	≈ 0	−0.46	−0.46	−0.64
Realistic recording, seven complete days	—	73.84	73.84	73.84
day 4 as missing minutes	—	68.75	68.75	none[a]
day 4 deleted	—	68.75[b]	71.94	35.18

[a] Fewer than the default minimum of comparison days remain, 66.20 with the minimum set to two days.
[b] Weekday-index pairing of the published implementation. A position-based implementation of the formula gives 71.94, the same as `GGIR`.

5 Input variants, calculator agreement and the association with BMI

Figure S2 shows the released classification at the epoch level, Figure S3 the distribution of the SRI under each input variant, and Figure S4 the agreement of `GGIR` and `sleepreg` with the Phillips formula on each input. Given the same timeline, `GGIR` and the Phillips formula differed by 0.08 points on average (SD of the difference 2.0; $r = 0.993$). `sleepreg` differed from the formula by 0.02 points at its default minimum of five comparison days (SD 0.3; $r = 0.9998$) and by 0.03 points at three days (SD 1.7; $r = 0.995$). The agreement was the same among the 60% of participants with seven consecutive recorded days and among the 40% with fewer than seven consecutive days ($r = 0.993$ and 0.992 for `GGIR`). `sleepreg` returned no value for 3412 of the 7085 participants at its default and for 1743 at three days.

On the released series the Phillips formula gave a median SRI of 61.3 (interquartile range 48.5 to 71.1). The rolling majority raised the median to 69.1, `GGIR`'s SRI1 smoothing to 69.7, the bridged series to 67.6 and the main sleep period alone to 75.5 (64.7 to 83.3). The mean number of sleep–wake transitions fell from 17.7 per day to between 2.1 and 4.5, whereas total sleep per day stayed between 7.2 and 8.6 hours, except under the 80% rule, which removed approximately 100 minutes of sleep per day. The smoothed scores remained correlated with the score on the unsmoothed series ($r = 0.97$ for the majority filter and the bridged series, 0.89 for the 80% rule, 0.75 for the main sleep period alone).

Table S4 gives the SRI–BMI multiplication factors for every input variant and calculator. In the fully adjusted model the factor was 0.955 (95% CI 0.936, 0.976) on the released series, 0.964 (95% CI 0.946, 0.983) after the majority filter, 0.969 (95% CI 0.952, 0.986) after `GGIR`'s smoothing, 0.963 (95% CI 0.945, 0.982) after bridging and 0.972 (95% CI 0.955, 0.989) for the main sleep period alone. In women the corresponding factors were 0.922, 0.933, 0.936, 0.932 and 0.946, in men 0.987 to 1.002. On every input the fully adjusted factor from `GGIR` differed from the factor of the formula by at most 0.002 and the factor for women by at most 0.005. The factors from `sleepreg` were 0.009 to 0.020 lower, i.e. the inverse association was somewhat stronger in its output. This difference is explained by the sample: at its default minimum of five comparison days `sleepreg` scored only participants with seven consecutive recorded days. The formula on the same 3673 participants gives the same fully adjusted factor as `sleepreg` (0.938 on the released series), and among the 60% of participants with seven consecutive days the association was stronger (fully adjusted factor 0.946 with the formula on the released series, `check_subgroups.R`). The ordering of the input variants was the same for every calculator, from the released series to the main sleep period alone. The sample on which `sleepreg` could estimate the association was half of the published sample at its default setting and three quarters at three days. The participants lost were those with a gap in the recording (29.5% of the sample) or with fewer than seven complete days (10.5%).

5.1 Model diagnostics

The fully adjusted survey-weighted model refitted with the SRI of each input variant and calculator satisfied the regression assumptions to the same degree as the published model (residual skewness and kurtosis, variance inflation, homoscedasticity and linearity diagnostics from `performance::`

`check_model()` and `car::qqPlot()`). The outputs are provided in the companion repository. The bootstrap intervals rest on resamples of participants drawn without regard to strata and primary sampling units, with the survey design rebuilt in each resample. Design-based intervals from Taylor linearisation (29 strata, 61 primary sampling units, 32 design degrees of freedom) for the fully adjusted factor were 0.950 (95% CI 0.932, 0.969) for the published SRI against the bootstrap 0.950 (95% CI 0.930, 0.969), 0.955 (95% CI 0.938, 0.973) against 0.955 (95% CI 0.936, 0.976) for the released series and 0.972 (95% CI 0.956, 0.989) against 0.972 (95% CI 0.955, 0.989) for the main-sleep-period input. In the random sample they were 0.952 (95% CI 0.908, 0.998) against 0.952 (95% CI 0.907, 0.996) for the released classification and 0.993 (95% CI 0.951, 1.037) against 0.993 (95% CI 0.951, 1.025) for SWaN. Over the fully adjusted and the sex-interaction models, the two kinds of interval differed at either end by at most 0.005 in the published sample and by at most 0.013 in the random sample (`check_taylor_intervals.R`).

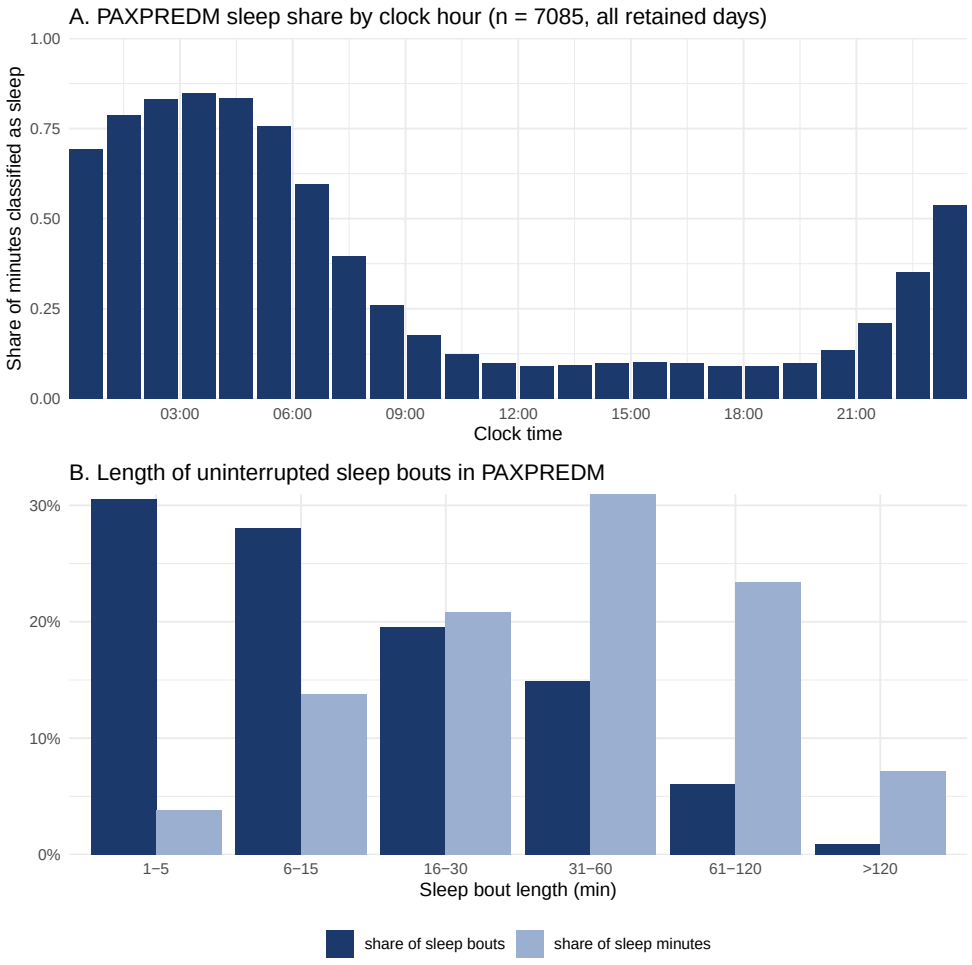


Figure S2: The released classification at the epoch level. (A) Share of minutes classified as sleep by clock hour, all retained days. (B) Length of uninterrupted sleep bouts, as a share of bouts and as a share of sleep minutes.

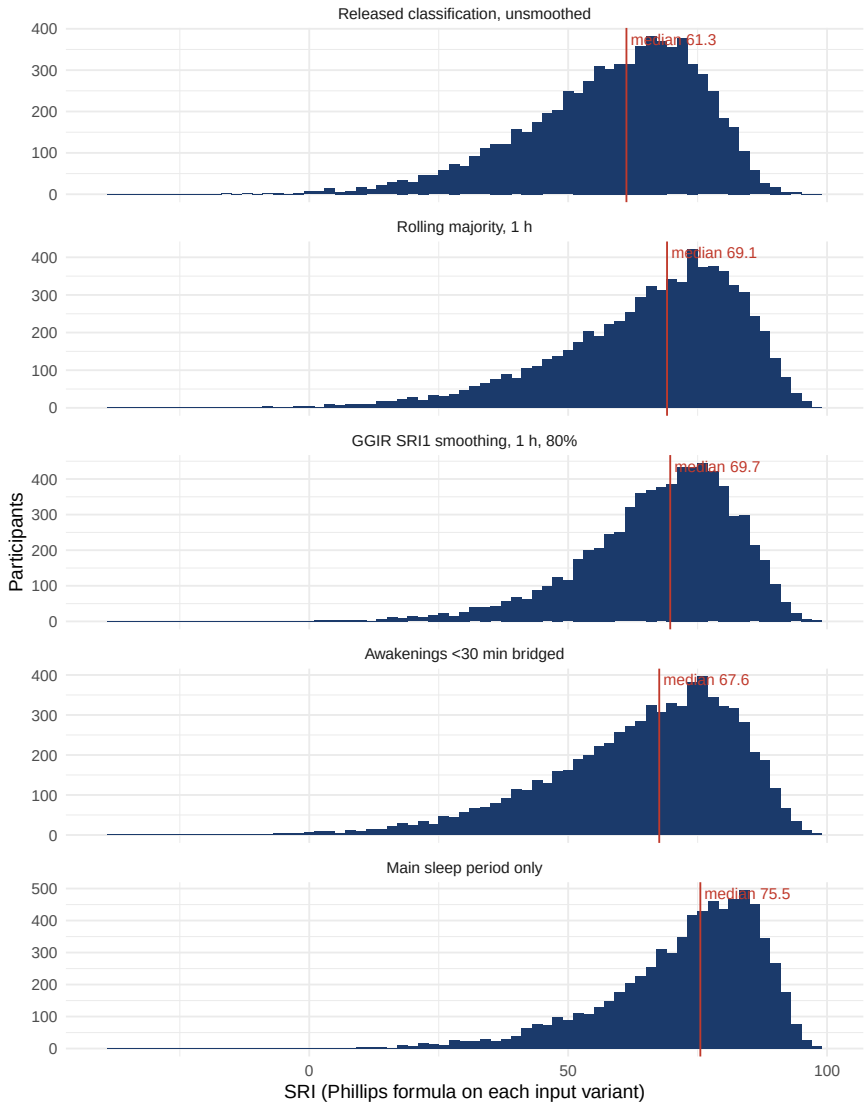


Figure S3: Distribution of the SRI (Phillips formula) under each input variant (7085 participants, of whom 7076 have a value), with medians.

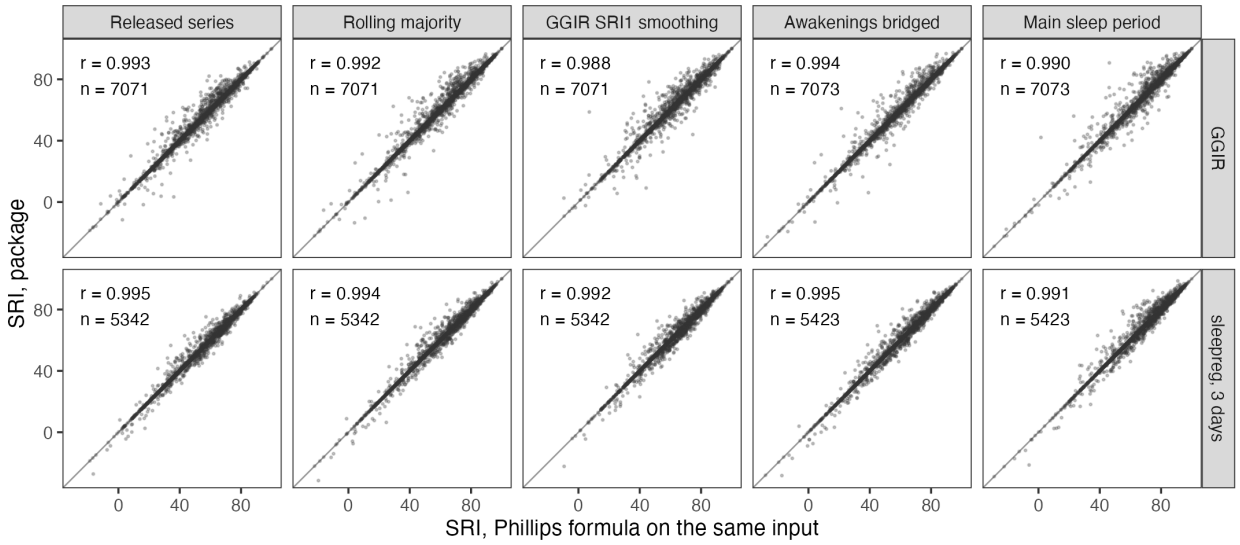


Figure S4: SRI from `GGIR` (top row) and from `sleepreg` with a minimum of three comparison days (bottom row) against the Phillips formula computed on the same input, for the released series and the four smoothed input variants (7085 participants, of whom 7076 have a formula value, and `sleepreg` returns no value for participants below its minimum number of days). The grey line is the identity. *r* is the Pearson correlation.

Table S4: SRI–BMI multiplication factor (top versus bottom SRI quintile) by input variant and calculator, computed with the method and models of the earlier study (Models 1 to 3 of its Table 3, survey-weighted, 95% bootstrap intervals). Model 1 adjusts for age and sex, Model 2 is fully adjusted and Model 3 adds a sex × SRI interaction and is shown for women and for men. The first row refits the published SRI values.

Input	Calculator	*n*	Model 1 (age, sex)	Model 2 (fully adjusted)	Model 3, women	Model 3, men
Published SRI	published implementation	7085	0.923 (0.907, 0.941)	0.950 (0.930, 0.969)	0.918 (0.891, 0.945)	0.982 (0.957, 1.005)
Released classification, unsmoothed	Phillips formula	7076	0.926 (0.911, 0.944)	0.955 (0.936, 0.976)	0.922 (0.902, 0.949)	0.987 (0.962, 1.013)
	`GGIR`	7071	0.926 (0.914, 0.942)	0.955 (0.940, 0.974)	0.921 (0.899, 0.941)	0.988 (0.970, 1.010)
	`sleepreg`, 5 days	3673	0.907 (0.886, 0.930)	0.938 (0.915, 0.964)	0.906 (0.874, 0.939)	0.976 (0.952, 1.009)
	`sleepreg`, 3 days	5342	0.916 (0.899, 0.932)	0.942 (0.923, 0.964)	0.910 (0.884, 0.933)	0.977 (0.956, 1.006)
Rolling majority, 1 h	Phillips formula	7076	0.934 (0.918, 0.951)	0.964 (0.946, 0.983)	0.933 (0.910, 0.958)	0.994 (0.969, 1.019)
	`GGIR`	7071	0.934 (0.921, 0.950)	0.964 (0.948, 0.982)	0.930 (0.907, 0.952)	0.995 (0.978, 1.017)
	`sleepreg`, 5 days	3673	0.914 (0.894, 0.934)	0.946 (0.926, 0.971)	0.917 (0.887, 0.948)	0.982 (0.959, 1.014)
	`sleepreg`, 3 days	5342	0.923 (0.907, 0.938)	0.950 (0.933, 0.972)	0.918 (0.893, 0.943)	0.984 (0.963, 1.010)
`GGIR` SRI1 smoothing, 1 h, 80%	Phillips formula	7076	0.940 (0.922, 0.956)	0.969 (0.952, 0.986)	0.936 (0.910, 0.963)	1.002 (0.980, 1.025)
	`GGIR`	7071	0.938 (0.925, 0.953)	0.967 (0.950, 0.982)	0.931 (0.909, 0.953)	1.000 (0.979, 1.021)
	`sleepreg`, 5 days	3673	0.922 (0.901, 0.947)	0.952 (0.929, 0.980)	0.921 (0.891, 0.956)	0.991 (0.969, 1.025)
	`sleepreg`, 3 days	5342	0.929 (0.911, 0.947)	0.955 (0.935, 0.975)	0.925 (0.899, 0.948)	0.987 (0.964, 1.011)
Awakenings <30 min bridged	Phillips formula	7076	0.933 (0.918, 0.950)	0.963 (0.945, 0.982)	0.932 (0.910, 0.959)	0.992 (0.968, 1.017)
	`GGIR`	7073	0.934 (0.917, 0.948)	0.964 (0.947, 0.979)	0.932 (0.910, 0.954)	0.994 (0.973, 1.014)
	`sleepreg`, 5 days	3854	0.912 (0.894, 0.935)	0.943 (0.922, 0.970)	0.916 (0.885, 0.943)	0.974 (0.947, 1.010)
	`sleepreg`, 3 days	5423	0.922 (0.904, 0.939)	0.949 (0.926, 0.965)	0.920 (0.893, 0.946)	0.980 (0.952, 1.005)
Main sleep period only	Phillips formula	7076	0.947 (0.930, 0.965)	0.972 (0.955, 0.989)	0.946 (0.923, 0.972)	0.996 (0.973, 1.018)
	`GGIR`	7073	0.948 (0.931, 0.959)	0.973 (0.958, 0.986)	0.945 (0.921, 0.967)	0.997 (0.976, 1.014)
	`sleepreg`, 5 days	3854	0.935 (0.918, 0.957)	0.963 (0.944, 0.988)	0.939 (0.910, 0.971)	0.990 (0.964, 1.017)
	`sleepreg`, 3 days	5423	0.938 (0.923, 0.954)	0.962 (0.943, 0.978)	0.938 (0.914, 0.963)	0.986 (0.961, 1.010)

6 The classifier behind PAXPREDM and the derivation settings

6.1 Identification of the classifier

The NHANES documentation does not name the algorithm behind PAXPREDM. It describes an open-source machine-learning classification of the 80 Hz signal into wake wear, sleep wear and non-wear in three steps [16]. In the first step, signal features are extracted from 1.5-minute windows and each 30-second window is classified, with a confidence value for each class. Stretches recorded in idle sleep mode (ISM) are replaced beforehand by the 10 seconds of signal that precede them, repeated over the stretch. In ISM the ActiGraph GT3X+ outputs a constant value during a period without movement until a threshold is exceeded on any axis [10]. The minute files record for every minute how many of its 4800 samples were taken in ISM (PAXAISMM). In the second step, periods shorter than a minimum duration (wake wear under 3 minutes, sleep wear and non-wear under 10 minutes) are reassigned on the basis of the confidence values of the adjacent periods. In the third step an orientation rule reassigns longer periods between non-wear and sleep. The 30-second resolution is explained by the polysomnography scoring of the validation data. This description corresponds step by step to the SWaN algorithm (sleep-wear, wake-wear and nonwear) of Thapa-Chhetry et al. [10], which was developed on ISM-filtered NHANES 2011–2014 recordings with polysomnography and directly observed wear as reference, validated against polysomnography and diaries, and which its authors state was used to generate the sleep summary measures of the survey. The data descriptor of a public derivative of the minute files attributes PAXPREDM to the same publication [15]. The public implementation (`swan_accel` 1.18, Python) is maintained by the first author of that publication. Whether the released classification was produced with this version of the code and with the ISM replacement described above is not documented.

6.2 Settings and minute labels

GGIR 3.3-6 was run as listed in Table S2. The 5-second epoch classes of part 5 were mapped to minutes as follows. A minute has twelve epochs. It is missing when more than half of them are flagged as non-wear, otherwise sleep when more than half of its twelve epochs are sleep, and wake otherwise. For GGIR SIB, sleep is the class `spt_sleep` (inside the detected sleep period and inside a sustained inactivity bout). For GGIR SPT, sleep is every epoch inside the detected sleep period. SWaN was run with `swan_first_pass` on each hourly file (feature extraction and classification of 30-second windows) and `swan_second_pass` on each day (minimum-duration and orientation rules). A minute took the common label of its two 30-second windows when they agreed and was unknown otherwise. Non-wear and unknown minutes are missing. The ISM stretches were used as recorded, without the replacement by the preceding 10 seconds. Minute k of a derived series (minutes counted from the first sample of the recording) was matched to the released minute with sample counter $\mathrm{PAXSSNMP}/4800 = k$. All series were restricted to the days retained in the rebuilt series.

7 Sample, computing platforms and stability of the derivation

7.1 Random order and sample

The 7085 participants of the published sample were placed in a random order with R's default generator and the seed 20261016 and their raw-signal archives were processed in that order. The script `make_random_order.R` and the file `analytic_random_order.tsv` in the companion repository hold the order. The fixed order allows interim analyses on the participants with the smallest ranks whose series are complete. The analyses in the main text use the first 1000 participants of the order. For 3 of them (NHANES identifiers 83452, 69334, 64226) SWaN did not finish in 60 minutes in two attempts and these participants were excluded, which leaves 997 participants with all four series.

7.2 Computing platforms and equality between platforms

Table S5 lists the platforms. The cloud machines ran a container built from `rocker/r-ver:4.6.1` with the same versions of `GGIR` and of all 109 R packages as the laptops and a Python 3.8 environment with the pinned SWaN dependencies. The container definition is in the companion repository.

R's `cumsum()` accumulates in the C type `long double`, which is 8 bytes on Apple processors and 16 bytes (80-bit extended precision) on x86-64. The autocalibration of `GGIR` (`g.calibrate`) computes rolling means of the 10-second window features with `cumsum()` and then removes windows whose features equal those of the preceding window, a rule that targets repeated constant signal such as the ISM stretches. The number of windows removed differed between the two precisions, the calibration coefficients differed in consequence, the calibrated accelerations differed by up to approximately 3% and the z-angle by up to 0.1 degrees. For one of six test participants the sleep/wake labels of some minutes differed (SRI 1.5 and 2.5 points higher for `GGIR` SIB and SPT on x86-64). SWaN was unaffected (identical classes, class probabilities that differed by at most 0.013). For the cloud runs the accumulation inside `g.calibrate` was therefore replaced by double-precision accumulation (a short patch loaded before `GGIR`, included in the repository), which reproduced the minute labels of the Apple laptop. Equality was verified on every participant processed both on the Apple M5 Max laptop and on another machine (15 on AWS, 26 on the two Hetzner server types, 23 on Vultr and 16 on the Apple M1 Pro laptop, which needs no patch): the sleep/wake minute labels of all three derived series were identical to the laptop's, the calibrated accelerations differed by at most 0.1 mg and the z-angle by at most 0.003 degrees, and SWaN returned identical classes with smoothed class probabilities that differed by up to 0.46 in single participants (`platform_check_labels.R`, results in `validate/platform_check/`).

Table S5: Computing platforms of the derivation (random sample, $n = 997$). Processing time is the sum over participants of the `GGIR` and SWaN run times (several participants were processed at a time on each machine). The median is per participant.

Platform	Participants	Processing time, h	Median, min
MacBook Pro, Apple M5 Max (18 cores, 36 GB), macOS	625	213	20
MacBook Pro, Apple M1 Pro (8 cores, 16 GB), macOS	27	11	25
AWS EC2 r7i (Intel Xeon, x86-64), Ubuntu container	249	119	29
Hetzner Cloud ccx33 and cpx62 (AMD EPYC, x86-64), Ubuntu container	70	29	25
Vultr Optimized Cloud Compute (AMD EPYC, x86-64), Ubuntu container	26	12	28
All platforms (15 October 2026 to 18 October 2026, 3.2 days)	997	384	22

7.3 Stability over the interim steps

Table S6 and Figure S5 compare the summaries after the first 500 and the first 1000 participants of the random order.

Table S6: Summaries of the derivation after the first 500 and the first 1000 participants of the random order (95% bootstrap intervals).

Statistic	Series	First 500 ($n = 497$)	First 1000 ($n = 997$)
Mean SRI	PAXPREDM	58.1 (56.5, 59.6)	58.1 (57.0, 59.2)
Mean SRI	GGIR SIB	69.6 (68.5, 70.6)	70.2 (69.5, 70.9)
Mean SRI	GGIR SPT	75.5 (74.2, 76.8)	75.8 (74.9, 76.7)
Mean SRI	SWaN	67.8 (66.3, 69.2)	68.3 (67.3, 69.2)
Mean difference to PAXPREDM	GGIR SIB	11.5 (10.4, 12.6)	12.1 (11.3, 12.9)
Mean difference to PAXPREDM	GGIR SPT	17.3 (16.3, 18.4)	17.7 (16.9, 18.5)
Mean difference to PAXPREDM	SWaN	9.7 (8.7, 10.5)	10.1 (9.5, 10.8)
Transitions per day	PAXPREDM	17.6 (17.1, 18.1)	17.8 (17.4, 18.2)
Transitions per day	GGIR SIB	19.9 (19.3, 20.5)	19.5 (19.1, 19.9)
Transitions per day	GGIR SPT	2.0 (2.0, 2.0)	2.0 (1.9, 2.0)
Transitions per day	SWaN	3.0 (2.9, 3.1)	3.0 (2.9, 3.0)
Agreement with PAXPREDM, %	GGIR SIB	88.2 (87.6, 88.8)	87.8 (87.3, 88.2)
Agreement with PAXPREDM, %	GGIR SPT	88.7 (88.0, 89.3)	88.2 (87.8, 88.7)
Agreement with PAXPREDM, %	SWaN	89.6 (89.1, 90.1)	89.4 (89.0, 89.7)

7.4 Association with BMI per SRI point and by sex

The quintile contrast was narrower for the derived series than for the released classification (median SRI difference between the top and bottom quintile 26.5 to 37.9 points against 42.9; Table 4 of the main text). The attenuation persists per SRI point: the BMI change associated with a 10-point higher SRI in the fully adjusted model was -1.1% (95% CI -2.2%, -0.1%) with the released classification and $+0.2\%$ (95% CI -1.7%, $+1.5\%$), 0.0% (95% CI -1.4%, $+1.1\%$) and -0.2% (95% CI -1.3%, $+0.7\%$) with GGIR SIB, GGIR SPT and SWaN. In the age- and sex-adjusted model the factors were 0.920 for the released classification and 0.957 to 0.969 for the derived series. In the model with a sex $\times$ SRI interaction the factors in women were 0.888 (95% CI 0.819, 0.951) with the released classification and 0.956 to 0.973 with the derived series. In men they were 1.014 (95% CI 0.957, 1.067) and 1.028 to 1.050.

7.5 Sensitivity analysis with each series' own missing minutes

In the sensitivity analysis each derived series kept its own non-wear and unknown minutes on the retained days. Participants with 30% or more missing minutes in a derived series were excluded (0.3% for the GGIR series, 0.0% for SWaN). The mean SRIs were then 69.3 for GGIR SIB, 75.4 for GGIR SPT and 67.5 for SWaN. Table S7 gives the SRI–BMI factors on the 994 remaining participants.

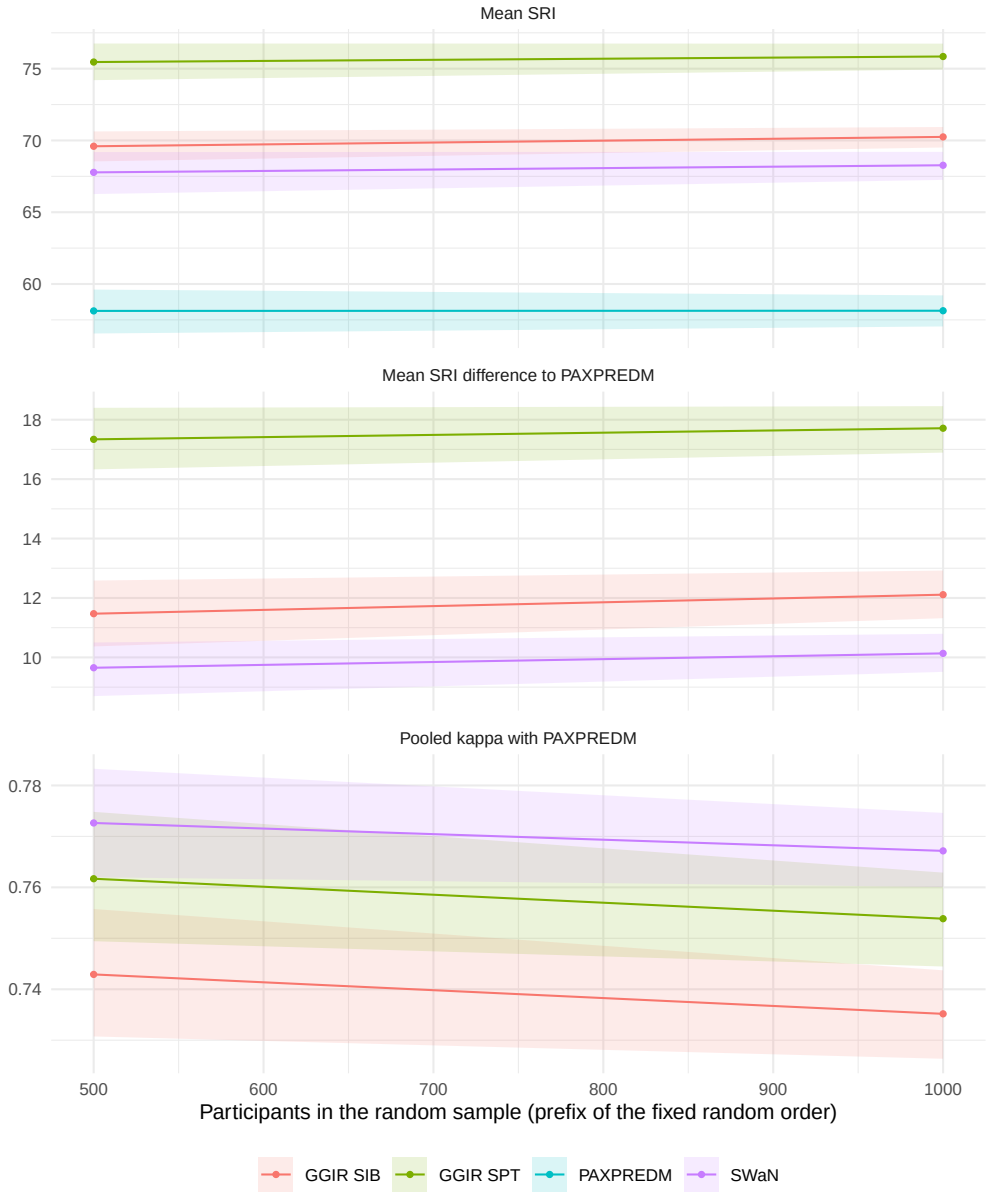


Figure S5: Mean SRI, mean difference from PAXPREDM and pooled kappa with PAXPREDM by the number of participants in the random sample (prefix of the fixed random order), with 95% bootstrap bands.

Table S7: SRI–BMI multiplication factor (top versus bottom SRI quintile) with each series' own missing minutes ($n = 994$; survey-weighted models; 95% bootstrap intervals). Models and ΔSRI (median SRI difference between the top and bottom quintile) as in Table 4 of the main text.

Series	ΔSRI	Model 1 (age, sex)	Model 2 (full adj.)	Model 3, women	Model 3, men	Ratio to PAXPREDM (Model 2)
PAXPREDM	43.0	0.920 (0.882, 0.965)	0.953 (0.909, 0.998)	0.887 (0.821, 0.936)	1.017 (0.959, 1.071)	—
GGIR SIB	25.8	0.974 (0.936, 1.013)	1.009 (0.969, 1.045)	0.975 (0.924, 1.033)	1.053 (0.994, 1.105)	1.058 (1.023, 1.102)
GGIR SPT	33.8	0.965 (0.927, 1.001)	1.001 (0.960, 1.036)	0.976 (0.925, 1.023)	1.028 (0.968, 1.082)	1.049 (1.016, 1.089)
SWaN	38.5	0.961 (0.924, 0.997)	0.997 (0.962, 1.032)	0.962 (0.906, 1.020)	1.033 (0.979, 1.079)	1.046 (1.015, 1.085)

8 Model 4: interactions with sex and race/ethnicity

Model 4 of the earlier study regresses log BMI on SRI with sex $\times$ SRI and race/ethnicity $\times$ SRI interactions and the covariates of Model 2, survey-weighted. For each race/ethnicity and sex group the multiplication factor for the top versus the bottom SRI quintile is $\exp(\beta_g \Delta)$, where β_g is the group's SRI slope and Δ the median SRI difference between the top and bottom quintile of the input. The intervals are percentile intervals from the same 200-resample participant bootstrap as for Models 1 to 3. The earlier study reported the bootstrap means of 100 resamples as point estimates. The refit of the published SRI differs from its Table 3 by at most 0.014 in the twelve groups. Table S8 gives the factors for the published SRI and for the Phillips formula on the released and the four smoothed inputs. On the same input, `GGIR` agreed with the formula to 0.013 or less in every group. `sleepreg` with its default of five days differed by up to 0.064 and with three days by up to 0.026, the largest differences in the smallest groups, because it returns a value for fewer participants (Table S4). The sex pattern held in every input: within each race/ethnicity group the factor of women was below that of men in 36 of 36 comparisons. Smoothing moved the group factors by -0.020 to $+0.030$.

Figures S6 and S7 repeat the earlier study's Figure 2 for the released series and for the main-sleep-period input: predicted BMI over the SRI range from 0 to 100 by race/ethnicity and sex, with the other covariates held at the published sample's means (continuous) and most frequent levels (categorical) and with 95% bands from the bootstrap coefficient vectors. The SRI range extends beyond the observed values at both ends, as in the earlier study.

Table S9 gives Model 4 on the random sample for the released classification and the three derived series. The groups hold 11 to 226 participants. The intervals are wide (widths 0.13 to 0.88). Women had a lower factor than men in all six race/ethnicity groups for each of the four series.

Table S8: Model 4 of the earlier study: SRI–BMI multiplication factor (top versus bottom SRI quintile) by race/ethnicity and sex, for the published SRI (published implementation) and for the Phillips formula on the released classification and the four smoothed inputs (published sample, survey-weighted, 95% bootstrap intervals). *n* is the group size in the published sample. The Phillips-formula columns rest on the 7076 participants with a value.

Race/ethnicity	Sex	*n*	Published SRI	Released, unsmoothed	Rolling majority	`GGIR` SRI1 smoothing	Bridged	Main sleep period
Mexican American	women	425	0.969 (0.923, 1.022)	0.967 (0.924, 1.016)	0.972 (0.932, 1.024)	0.981 (0.942, 1.031)	0.974 (0.934, 1.024)	0.990 (0.949, 1.034)
	men	401	1.035 (0.989, 1.091)	1.034 (0.994, 1.080)	1.035 (0.995, 1.085)	1.048 (1.010, 1.096)	1.035 (0.991, 1.085)	1.042 (1.010, 1.087)
Other Hispanic	women	380	0.930 (0.884, 0.985)	0.939 (0.899, 0.991)	0.946 (0.899, 1.000)	0.949 (0.911, 1.004)	0.949 (0.908, 1.001)	0.959 (0.913, 1.008)
	men	302	0.994 (0.948, 1.037)	1.003 (0.959, 1.056)	1.007 (0.964, 1.056)	1.014 (0.969, 1.064)	1.009 (0.969, 1.057)	1.009 (0.967, 1.052)
Non-Hispanic White	women	1536	0.906 (0.877, 0.935)	0.912 (0.887, 0.945)	0.922 (0.896, 0.950)	0.924 (0.896, 0.955)	0.922 (0.896, 0.948)	0.934 (0.906, 0.963)
	men	1372	0.968 (0.942, 0.993)	0.975 (0.945, 1.001)	0.981 (0.952, 1.009)	0.987 (0.960, 1.015)	0.980 (0.949, 1.008)	0.983 (0.957, 1.006)
Non-Hispanic Black	women	872	0.970 (0.938, 1.005)	0.968 (0.937, 1.008)	0.979 (0.945, 1.020)	0.996 (0.963, 1.032)	0.973 (0.939, 1.013)	0.995 (0.965, 1.031)
	men	787	1.036 (0.997, 1.069)	1.035 (0.998, 1.073)	1.042 (1.007, 1.079)	1.064 (1.032, 1.101)	1.034 (1.000, 1.069)	1.047 (1.016, 1.087)
Non-Hispanic Asian	women	428	0.925 (0.891, 0.959)	0.923 (0.888, 0.960)	0.935 (0.901, 0.973)	0.930 (0.897, 0.974)	0.939 (0.904, 0.977)	0.947 (0.910, 0.984)
	men	369	0.988 (0.944, 1.031)	0.986 (0.949, 1.029)	0.995 (0.957, 1.044)	0.993 (0.956, 1.039)	0.999 (0.959, 1.046)	0.997 (0.962, 1.039)
Other, incl. multiracial	women	111	0.868 (0.791, 0.964)	0.887 (0.797, 0.988)	0.900 (0.803, 1.010)	0.905 (0.816, 1.012)	0.895 (0.808, 0.999)	0.882 (0.785, 1.000)
	men	102	0.927 (0.841, 1.036)	0.948 (0.853, 1.055)	0.958 (0.855, 1.080)	0.966 (0.869, 1.096)	0.951 (0.862, 1.063)	0.928 (0.823, 1.070)

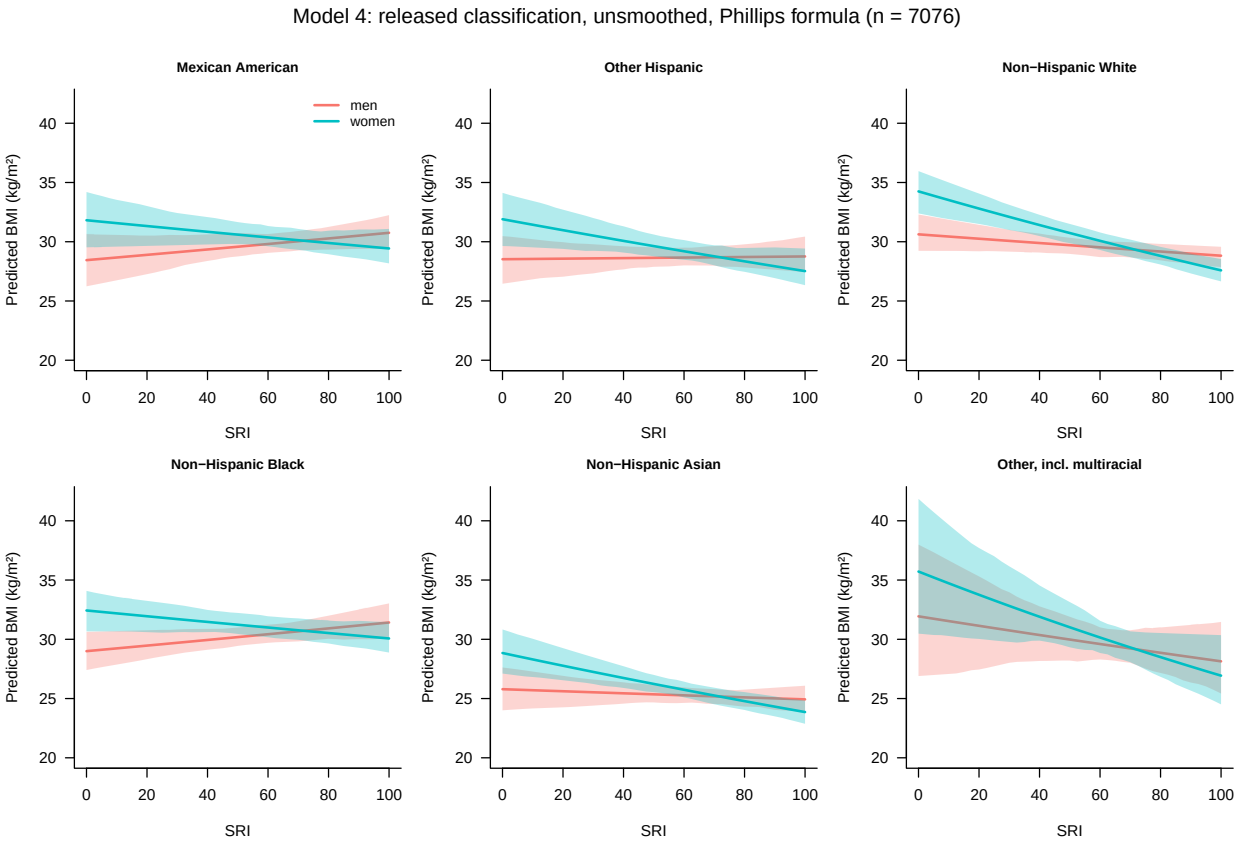


Figure S6: Predicted BMI over the SRI range from Model 4 on the released classification (Phillips formula, $n = 7076$), by race/ethnicity and sex, with the other covariates at the published sample's means and most frequent levels. Shaded bands are 95% bootstrap bands. The layout follows Figure 2 of the earlier study.

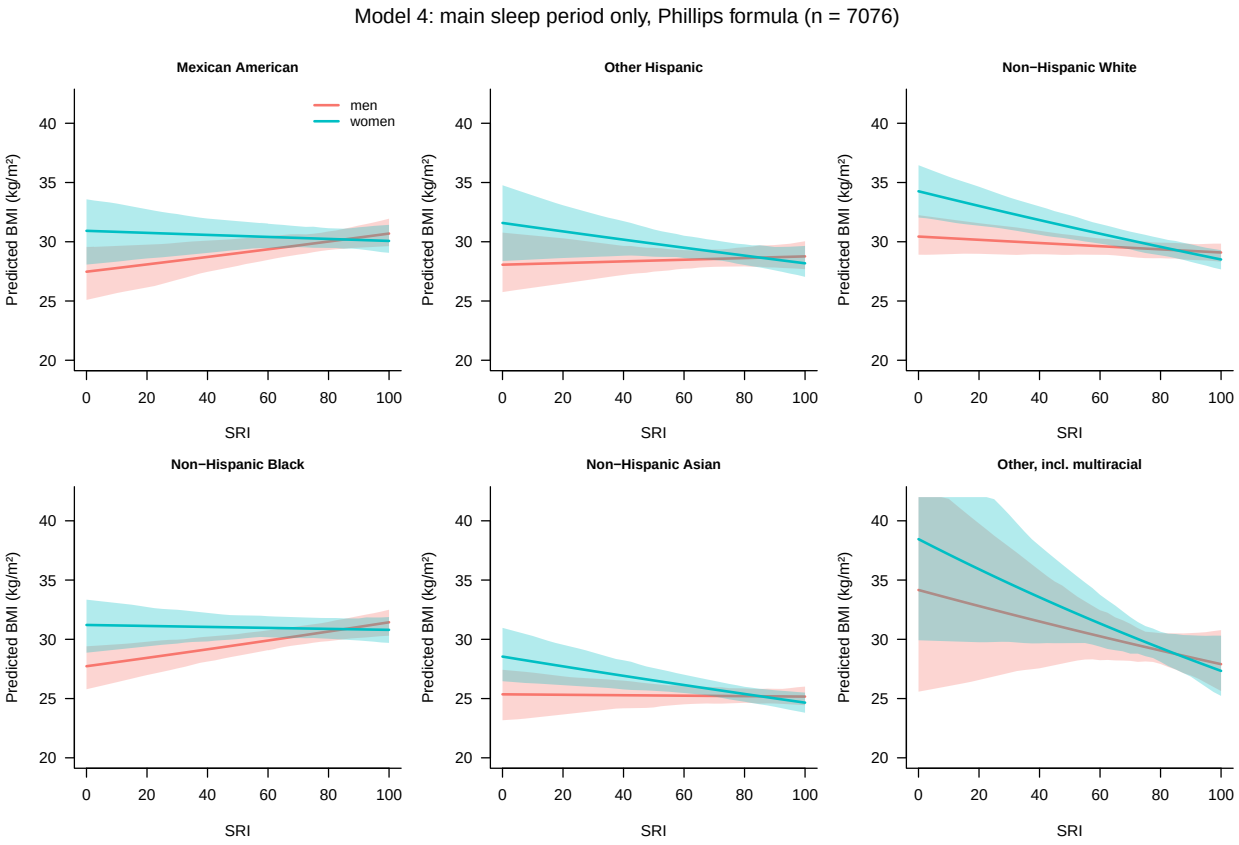


Figure S7: As Figure S6, for the main-sleep-period input (Phillips formula, $n = 7076$).

Table S9: Model 4 on the random sample ($n = 997$): SRI–BMI multiplication factor (top versus bottom SRI quintile) by race/ethnicity and sex for the released classification and the three series derived from the raw signal (survey-weighted; 95% bootstrap intervals). n is the number of participants in the group.

Race/ethnicity	Sex	n	PAXPREDM	GGIR SIB	GGIR SPT	SWaN
Mexican American	women	50	0.999 (0.880, 1.138)	1.091 (0.940, 1.285)	1.076 (0.913, 1.250)	1.066 (0.935, 1.211)
	men	54	1.133 (1.008, 1.295)	1.175 (1.019, 1.357)	1.128 (0.979, 1.294)	1.137 (1.011, 1.285)
Other Hispanic	women	60	0.956 (0.831, 1.109)	0.933 (0.847, 1.054)	0.930 (0.817, 1.049)	0.973 (0.860, 1.120)
	men	41	1.083 (0.937, 1.217)	1.005 (0.886, 1.141)	0.974 (0.857, 1.104)	1.038 (0.894, 1.202)
Non-Hispanic White	women	226	0.867 (0.793, 0.939)	0.956 (0.874, 1.015)	0.965 (0.882, 1.025)	0.937 (0.867, 1.000)
	men	184	0.982 (0.906, 1.045)	1.030 (0.943, 1.092)	1.011 (0.926, 1.067)	0.999 (0.922, 1.049)
Non-Hispanic Black	women	134	0.932 (0.832, 1.023)	1.034 (0.964, 1.099)	1.042 (0.965, 1.122)	1.022 (0.949, 1.097)
	men	102	1.056 (0.963, 1.154)	1.113 (1.039, 1.185)	1.092 (1.022, 1.158)	1.090 (1.008, 1.169)
Non-Hispanic Asian	women	69	0.916 (0.825, 1.028)	0.964 (0.875, 1.040)	0.968 (0.879, 1.040)	0.960 (0.871, 1.049)
	men	55	1.038 (0.934, 1.165)	1.038 (0.927, 1.125)	1.015 (0.907, 1.107)	1.024 (0.908, 1.133)
Other, incl. multiracial	women	11	0.958 (0.651, 1.411)	0.822 (0.647, 1.142)	0.858 (0.650, 1.239)	0.942 (0.672, 1.247)
	men	11	1.085 (0.737, 1.614)	0.886 (0.702, 1.239)	0.899 (0.691, 1.325)	1.004 (0.724, 1.329)